

PSEM — Prediction State Establishment Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

Prediction Formation Standard (PFOS)

Operational Layer: Prediction Formation Governance Layer

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Prediction State Establishment

Version: 1.0

Status: Canonical · Open Module

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Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS™ · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · CLIA® · SIMULOS®

AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

Prediction State Establishment

Canonical Definition

Prediction State Establishment Module (PSEM) defines the governance conditions under which a prediction that has completed composition, Predictive Claim identification, and Prediction Output Formation is assigned a governed Formation State reflecting whether it is sufficiently formed, conditionally formed, restricted, incomplete, conflicting, or unable to proceed.

PSEM determines the current legitimacy state of the prediction at the end of Prediction Formation.



transforms:

Governed Prediction Composition + Identified Predictive Claim + Governed Prediction Output + Formation Governance Conditions

into:

Governed Prediction Formation State

PSEM does not determine whether the prediction is true, probable, feasible, or later reliable.

It determines whether the prediction has been sufficiently formed to proceed into subsequent prediction governance.

Its foundational distinction is:

Prediction Output Exists $\neq$ Prediction Fully Formed

Operational Role

PSEM serves as the fourth internal governance space of the Prediction Formation Standard (PFOS) .

It receives:

Governed Prediction Composition

Identified Predictive Claim

Governed Prediction Output

Unresolved Formation Gaps / Conditions / Restrictions

and determines the governed state of the prediction at Formation Time.

The module asks:

Has a sufficiently governed prediction actually been formed, and if not, what state most accurately represents its current formation condition?

The governed progression is:

Formed Components → Formation Sufficiency Assessment → Gap Review → Conflict Review → Conditionality Review → Restriction Assessment → Formation State Determination → Governed Prediction Formation State

The primary output is:

Governed Prediction Formation State

which becomes a principal input to:

PFTM — Prediction Formation Traceability Module.

Internal Module Flow

Prediction Composition → Predictive Claim Identification → Prediction Output Formation → Prediction State Establishment → Prediction Formation Traceability

Current Position: Prediction State Establishment

Module Operational Space

PSEM governs:

formation-state determination; formation sufficiency; formation completeness;

prediction readiness; Draft state; Formed state; Conditionally Formed state; Restricted state; Incomplete state; Conflicting state; Unable-to-Form state; state criteria; state transition; state justification; state restriction; unresolved formation gap; unresolved component conflict; unresolved claim ambiguity;

unresolved output conflict; assumption-related formation condition; methodology-related formation condition; basis-related formation condition; boundary-related formation condition; horizon-related formation condition; provenance-related formation condition; formation approval where applicable; formation rejection where applicable; progression eligibility; progression restriction; state version; and historical formation-state integrity.

Module Function

A prediction may have:

a defined Prediction Object; a documented methodology; sufficient Prediction Basis; governed assumptions; an identified Predictive Claim; an output.

Yet unresolved governance conditions may still prevent it from being fully formed.

Examples include:

a critical assumption remains unresolved; the claim contains material ambiguity; two candidate outputs conflict; Formation Time cannot be established; the Prediction Basis is conditionally sufficient; the methodology is only valid under a condition not yet confirmed.

PSEM prevents such a prediction from being represented simply as:

Formed

when a more accurate state is:

Conditionally Formed

Restricted

Incomplete

or:

Unable to Form.

Prediction Formation State

A Prediction Formation State is the governed status assigned to a prediction based on the sufficiency and condition of its formation process.

The state communicates whether the prediction:

exists only as a draft; has been sufficiently formed; is formed subject to explicit conditions; is restricted by unresolved governance gaps; remains incomplete; contains unresolved conflict; cannot legitimately be formed.

State ≠ Predictive Quality

Formation State does not determine whether the prediction is good or accurate.

A fully formed prediction may later be wrong.

A restricted prediction may later happen to be correct.

Formation State ≠ Predictive Accuracy

Draft State

Draft represents a prediction still under development and not yet established as a final Governed Prediction.

A Draft may contain:

candidate claim; candidate output; unresolved methodology elements; provisional assumptions; incomplete composition.

Draft should remain distinguishable from Formed.

Draft ≠ Formed

Formed State

Formed represents a prediction that meets the applicable formation requirements and contains no unresolved material Formation Gap that prevents progression.

A Formed prediction should possess sufficient:

composition coherence; claim identification; output formation; Prediction Object association; Purpose association; Horizon association; Boundary association where applicable; Methodology association; Prediction Basis association;

Assumption association; conditionality; Formation Time; provenance; reconstructability.

Formed ≠ Validated Future Outcome

A prediction can be Formed while remaining:

uncertain; low probability; difficult to achieve; later incorrect.

Formed Prediction ≠ Correct Prediction

Conditionally Formed State

Conditionally Formed represents a prediction that can legitimately exist but only subject to explicit unresolved or continuing conditions.

Examples:

an important assumption is Conditionally Holding; one material information source is provisional; methodology applicability depends upon a defined regime remaining stable; the claim is valid only under Scenario A.

Conditional Formation is legitimate only when the condition is visible and governable.

Conditionally Formed ≠ Unconditionally Formed

Conditional Formation Condition

A Conditional Formation Condition should identify:

the unresolved or continuing condition; why the prediction may still proceed; what consequence the condition may have; what trigger requires reassessment.

Restricted State

Restricted represents a prediction that has been materially formed but whose use, progression, interpretation, or reliance must remain limited because of a known Formation Governance Gap.

Restriction may arise from:

incomplete Prediction Basis; unresolved assumption uncertainty; material claim ambiguity; provenance limitation; incomplete boundary; unresolved output transformation; methodology limitation.

Restricted ≠ Invalid

A Restricted prediction may still possess analytical value.

Its governed use is simply limited.

Restricted Prediction $\neq$ Invalid Prediction by Default

Incomplete State

Incomplete represents a prediction formation process where material required elements are still missing.

Examples:

no final Predictive Claim; missing Horizon; missing Output State; missing material assumption relationship; missing methodology reference.

An incomplete prediction should not proceed as fully formed.

Conflicting State

Conflicting represents a formation condition where material unresolved conflicts remain within the prediction structure.

These may include:

conflicting outputs; conflicting claims; conflicting assumptions; incompatible boundaries; incompatible time horizons; competing formation states.

A prediction may remain Conflicting until the conflict is:

resolved; separated; qualified; or causes inability to form.

Unable-to-Form State

Unable to Form represents a governance conclusion that available predictive components cannot legitimately establish a Governed Prediction.

Possible reasons include:

undefined Prediction Object; materially insufficient methodology; insufficient Prediction Basis; ungovernable critical assumption; irreconcilable claim ambiguity; irreconcilable output conflict; inability to establish Formation Time; severe provenance failure; fundamentally incompatible components.

Unable to Form $\neq$ Prediction of Non-Occurrence

It is a governance state.

State Determination Criteria

Formation State should be determined using explicit criteria.

These may include:

component coherence; claim clarity; output integrity; methodology sufficiency; Basis sufficiency; assumption governance sufficiency; Horizon definition; Boundary definition; conditionality visibility; provenance sufficiency; traceability readiness; unresolved gap severity.

Formation Sufficiency

Formation Sufficiency represents whether the assembled predictive structure contains enough governed information and coherence to justify a Formed state.

Sufficiency is contextual.

It should reflect Prediction Purpose and governed risk.

Formation Completeness

Completeness concerns whether materially required formation elements exist.

Completeness does not require maximum documentation.

More Documentation ≠ More Complete Formation by Default

Formation Readiness

Formation Readiness represents whether the prediction is ready to proceed into subsequent PGA™ governance spaces.

A prediction may be:

Ready

Conditionally Ready

Restricted

Not Ready

Progression Eligibility

PSEM may determine whether a prediction can proceed to:

PPUS — Prediction Probability & Uncertainty Standard

and then onward through the Prediction Lifecycle.

Potential states may include:

Proceed

Proceed Conditionally

Proceed with Restriction

Do Not Proceed

Progression Eligibility ≠ Reliance Permission

A prediction may be eligible to continue through governance without yet being permitted for operational reliance.

PCRS later governs reliance.

Formation Gap Review

PSEM receives unresolved gaps from PCOM, PCIM, and POFM.

These may include:

Composition Gap; Claim Identification Gap; Output Formation Gap.

PSEM determines whether those gaps:

prevent formation; permit conditional formation; require restriction; remain minor.

Material Formation Gap

A Material Formation Gap is a gap capable of materially altering:

claim meaning; output meaning; methodology applicability; Prediction Basis sufficiency; assumption interpretation; Formation-Time reconstruction; downstream governance.

Material gaps must not be hidden inside a Formed state.

Critical Formation Gap

A Critical Formation Gap prevents legitimate Prediction Formation.

Examples may include:

no identifiable Predictive Claim; no valid Formation Time; irreconcilable semantic conflict; no methodologically supportable output.

Minor Formation Gap

A Minor Formation Gap may not prevent formation if it does not materially alter interpretation or governance.

It should still remain documented where relevant.

Gap Severity $\neq$ Number of Gaps

One Critical Formation Gap may be more significant than ten minor gaps.

Number of Gaps $\neq$ Formation Severity

Unresolved Assumption Condition

PAS may deliver assumptions in states such as:

Uncertain

Under Challenge

Conditionally Holding

PSEM determines whether those states:

permit full formation; require conditional formation; require restriction; prevent formation.

PSEM does not re-govern the assumption itself.

Critical Assumption and Formation State

An unresolved Critical Assumption may materially affect Formation State.

However:

Critical Assumption Exists $\neq$ Unable to Form by Default

The consequence depends upon its actual state and the prediction context.

Methodology Condition

PMES may establish methodological limitations or conditions.

If those conditions are unresolved at Formation Time, PSEM must reflect them in the prediction state.

Prediction Basis Condition

PBAS may establish a basis limitation.

PSEM determines the formation consequence of that limitation without taking over Prediction Basis Governance.

Claim Condition

A claim may be:

fully identified; conditionally identified; ambiguous; conflicting.

PSEM incorporates this state into formation-state determination.

Output Condition

A Prediction Output may be:

formed; conditionally formed; restricted; conflicting; incomplete.

PSEM determines the overall prediction state from the complete formation context.

Component State ≠ Overall Prediction State

A prediction's overall state must not be mechanically copied from any single component.

Example:

one input may be uncertain while the overall prediction remains Formed.

Conversely:

several individually acceptable components may combine into a Restricted or Conflicting prediction.

Formation State Composition

The overall Formation State should consider the combined state of:

Composition + Claim + Output + Methodology + Basis + Assumptions + Conditions + Traceability
Readiness

State Escalation

A prediction may move from:

Draft → Formed

or:

Draft → Conditionally Formed

or:

Conditionally Formed → Restricted

depending upon emerging governance conditions.

PSEM governs state change within the formation stage.

State De-escalation

A previously stronger Formation State may be reduced if a pre-progression issue is discovered before the prediction leaves the formation stage.

Example:

Formed → Restricted

because a critical provenance issue is discovered immediately after formation.

Formation-State Change ≠ Prediction Revision

A state correction within the same formation event does not necessarily constitute prediction-level revision.

PMRS owns revision governance later in the lifecycle.

Formation Approval

Some organizations may require formal approval before a prediction receives a Formed state.

PSEM can preserve:

approver reference; approval time; authority reference.

It does not create organizational authority.

GOA™ and AGA™ retain that space.

Formation Rejection

An authorized process may reject formation.

Rejection should preserve:

reason; state; unresolved gap; time.

Rejected Formation ≠ Prediction False

Formation State and Probability Governance

A prediction may proceed from PSEM into PPUS.

PSEM does not determine:

probability; confidence; uncertainty.

Formation State ≠ Probability State

Formation State and Feasibility

A Formed prediction may later be assessed as infeasible under PFES.

Formed ≠ Feasible

Formation State and Communication

A Formed prediction may still not be permitted for public or operational communication.

PCRS owns communication and reliance.

Formation State and Monitoring

Once a prediction enters the active lifecycle, later state changes caused by new information belong primarily to PMRS and related downstream standards.

PSEM governs state at formation.

Formation-State Version

Any material change in formation-state determination should remain historically identifiable.

Examples:

Draft → Conditionally Formed → Formed

or:

Draft → Restricted → Unable to Form

Historical Formation-State Integrity

Later lifecycle states must not overwrite the state that existed at Formation Time.

Current Prediction State ≠ Formation-Time Prediction State

State Justification

Every material Formation State should have a governed justification.

Examples:

Formed because all material formation requirements satisfied.

Conditionally Formed because Critical Assumption A remains Conditionally Holding.

Restricted because Prediction Boundary remains partially unresolved.

State Transparency

Users and systems consuming the prediction should be able to identify the Formation State where material.

A restricted or conditional prediction must not silently appear as fully formed.

Formation State and Automation

In automated environments, Formation State may control whether downstream systems are permitted to: continue processing; calculate probability; assess feasibility; publish; act; escalate.

PSEM can therefore operate as a machine-readable governance gate.

Machine-Readable State

PSEM may support canonical machine-readable states such as:

DRAFT

FORMED

CONDITIONALLY_FORMED

RESTRICTED

INCOMPLETE

CONFLICTING

UNABLE_TO_FORM

Implementation may use equivalent controlled states.

State Transition Control

Machine systems should not move a prediction from one Formation State to another without an identifiable trigger or governance condition.

State Inheritance

A downstream system must not interpret:

Conditionally Formed

as:

Formed

merely because its interface supports only a binary valid/invalid state.

State Simplification $\neq$ State Equivalence

Formation State Governance Sufficiency

Prediction State Establishment is sufficiently governed when:

explicit state criteria exist; formation gaps are reviewed; critical conflicts remain visible; conditionality remains visible; restriction rationale is explicit; inability to form is permitted; overall state is not mechanically inherited from one component; progression eligibility is governed;

state transitions are traceable; historical state remains reconstructable.

Governance Requirements

PSEM requires that:

1. every prediction completing Prediction Formation receive an identifiable Formation State; 2. Draft remain distinguishable from Formed; 3. Conditional Formation remain distinguishable from unrestricted formation; 4. Restricted remain distinguishable from Invalid; 5. Incomplete remain distinguishable from Unable to Form; 6. unresolved material conflicts remain visible; 7. Critical Formation Gaps prevent unrestricted Formed status; 8. Formation State reflect the combined formation context; 9. Component State not automatically determine overall Prediction State; 10. assumption conditions be considered where material; 11. methodology conditions be considered where material; 12. Prediction Basis limitations be considered where material; 13. claim ambiguity be considered where material; 14. output conflict be considered where material; 15. gap severity remain distinguishable from gap count; 16. inability to form remain a legitimate governance result; 17. Formation State not imply predictive correctness; 18. Formation State not imply feasibility; 19. Formation State not imply reliance permission; 20. progression eligibility remain explicit where material; 21. conditional or restricted states remain machine-readable where automation depends upon them; 22. State Simplification not erase material governance meaning; 23. state transitions remain traceable; 24. material state changes remain versioned or historically preserved; 25. Formation-Time state remain reconstructable.

Governance Logic

PSEM applies the following governance logic:

Governed Prediction Composition

Identified Predictive Claim

Governed Prediction Output

Formation Gaps / Conditions / Restrictions

↓

Formation Sufficiency Assessed

↓

Composition State Reviewed

↓

Claim State Reviewed

↓

Output State Reviewed

↓

Methodology / Basis / Assumption Conditions Reviewed

↓

Material / Critical Formation Gaps Classified

↓

Conditionality / Restriction Requirements Determined

↓

Progression Eligibility Determined

↓

Prediction Formation State Established

↓

State Justification Preserved

↓

Historical Formation State Preserved

↓

Governed Prediction Formation State

↓

PFTM — Prediction Formation Traceability Module

Possible outcomes:

FORMED

or:

CONDITIONALLY FORMED

or:

RESTRICTED

or:

INCOMPLETE

or:

CONFLICTING

or:

UNABLE TO FORM

Minimum Implementation Framework

A minimum implementation of PSEM must establish:

A. Prediction Reference

The prediction undergoing Formation State establishment.

B. Composition State Reference

The applicable state from PCOM.

C. Predictive Claim State Reference

The applicable state from PCIM.

D. Prediction Output State Reference

The applicable state from POFM.

E. Formation Gap Register

All material unresolved formation gaps.

F. Gap Severity

At minimum:

Minor / Material / Critical / Unable to Determine

G. Conditional Formation Reference

Any condition required for the prediction to proceed.

H. Restriction Reference

Any limitation on progression, interpretation, or downstream use.

I. Formation State

At minimum:

Draft / Formed / Conditionally Formed / Restricted / Incomplete / Conflicting / Unable to Form

J. State Justification

The governance basis for the assigned Formation State.

K. Progression Eligibility

At minimum:

Proceed / Proceed Conditionally / Proceed with Restriction / Do Not Proceed

L. State Trigger / Transition Reference

Any event or governance condition causing state transition.

M. Authority / Approval Reference

Where required by the applicable governance environment.

N. Formation-State Version

The applicable state version or equivalent historical identifier.

O. Historical Formation State

The Formation State applicable at the relevant Formation Time.

Governance Outputs

PSEM may produce:

Governed Prediction Formation State; Formation State Record; formation-sufficiency assessment; Formation Gap Register; Formation Gap Severity; conditional-formation record; restriction record; incomplete-formation record; conflict-state record; Unable-to-Form Record; state-justification record; progression-eligibility record; formation-approval reference; formation-rejection reference; state-transition record; Formation-State Version; machine-readable Formation State; historical Formation-State Record.

The primary governed output is:

Governed Prediction Formation State

which becomes a principal input to:

PFTM — Prediction Formation Traceability Module.

Critical Distinctions

Prediction Output Exists ≠ Prediction Fully Formed

Formation State ≠ Predictive Accuracy

Draft ≠ Formed

Formed Prediction ≠ Correct Prediction

Conditionally Formed ≠ Unconditionally Formed

Restricted Prediction ≠ Invalid Prediction by Default
Unable to Form ≠ Prediction of Non-Occurrence
More Documentation ≠ More Complete Formation by Default
Progression Eligibility ≠ Reliance Permission
Number of Gaps ≠ Formation Severity
Critical Assumption Exists ≠ Unable to Form by Default
Component State ≠ Overall Prediction State
Formation-State Change ≠ Prediction Revision
Rejected Formation ≠ Prediction False
Formation State ≠ Probability State
Formed ≠ Feasible
Current Prediction State ≠ Formation-Time Prediction State
State Simplification ≠ State Equivalence

Enterprise Application

PSEM gives organizations a controlled way to distinguish predictions that are:
fully formed; conditionally usable; restricted; incomplete; internally conflicting; unable to proceed.
Without Prediction State Establishment, organizations often collapse all predictive outputs into a binary:
prediction exists / prediction does not exist.

That is too weak for complex predictive systems.

A prediction may exist while still carrying governance conditions that materially affect whether downstream systems should:

process it; display it; communicate it; rely upon it; automate decisions from it.

PSEM creates a governed formation gate between output creation and the rest of the Prediction Lifecycle.

Use Case 1 — The AI Has Produced a Forecast, but One Critical Assumption Is Still Unresolved

An AI planning system predicts:

Production output: 52,000 units during Q4.

The Predictive Claim is clear and the Prediction Output is properly formed.

However, the prediction depends on a Critical Assumption:

The sole supplier of Component X will maintain at least 80% contracted capacity throughout Q4.

At Formation Time, the supplier is undergoing restructuring and future capacity cannot yet be determined.

The technical system marks the prediction simply as:

Complete.

The critical governance question is:

Does the existence of a complete model output justify treating the prediction as fully formed when a Critical Assumption remains unresolved?

PSEM evaluates the complete formation context and may establish:

Formation State: Conditionally Formed

with:

Condition: Supplier capacity remains sufficiently available

and a corresponding reassessment trigger.

The prediction exists.

Its conditional state remains visible.

Prediction Output Exists $\neq$ Prediction Fully Formed

Use Case 2 — An Automated System Treats “Restricted” as “Valid” and Executes Anyway

A financial prediction is created for internal scenario analysis.

Its methodology and output are coherent, but the Prediction Basis contains a material unresolved limitation.

PSEM establishes:

Formation State: Restricted

and:

Progression Eligibility: Proceed with Restriction

The prediction is permitted to continue for analytical evaluation but not yet for automated capital allocation.

A downstream trading system supports only:

Valid / Invalid

and converts:

Restricted $\rightarrow$ Valid.

The system automatically executes a trade.

The critical governance question is:

Can a downstream system legitimately collapse a governed formation state into a simpler state when that simplification removes the restriction that gives the prediction its actual governance meaning?

PSEM requires the material Formation State to remain preserved through downstream processing.

State Simplification $\neq$ State Equivalence

Canonical Boundary

PSEM governs the assessment, determination, justification, restriction, conditionality, progression eligibility, transition, versioning, machine-readable representation, and historical preservation of the Prediction Formation State. It does not govern Prediction Composition itself, Predictive Claim Identification itself, Prediction Output Formation itself, Prediction Methodology, Prediction Basis, Predictive Assumptions, probability and uncertainty determination, final Prediction Feasibility, communication and reliance, prediction-level revision, outcome determination, or predictive performance assessment.

PSEM establishes:

whether and in what governed state the prediction has been formed.

It does not establish:

whether the prediction is true, probable, feasible, or appropriate for final reliance.

Canonical Closing Statement

A prediction does not become fully governed merely because its pipeline has produced an output.

PSEM makes the formation condition explicit so that complete, conditional, restricted, conflicting, and incomplete predictions cannot silently become equivalent states.

The state of a prediction is part of the prediction's governance — not an implementation detail.

[↑](#)

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